

Domain name 101

Understanding domains and their hierarchy is an interesting way to analyze the Internet highways that we knowingly or unknowingly tread on a daily basis. We also take an aim at busting some common domain jargon



TLD

TLD simply expands to Top-Level Domain. To understand what a TLD is, you should first know the basic parts called labels. Let's take the example of the following web address, www.thinkdigit.com. In this case, 'www', 'thinkdigit' and '.com' are three distinct labels of a domain name. Here the '.com' bit is a top-level domain, while the part 'thinkdigit' (pointing to a working website) is called a hostname and 'www' is a subdomain of 'thinkdigit'.

A top-level domain constitutes one of the domains at the highest level in the working of the Internet's hierarchy of domain name system. The most prominent example of a TLD is the '.com' domain.

There are different types of TLDs as defined by ICANN (Internet Corporation for Assigned Names & Numbers), the governing body of all things related to Internet domain names. Let's take a closer look at them.

gTLD

A gTLD or generic Top-Level Domain is a sub-set of TLDs that don't have any specific geographic relevance to their audience, neither do they have any affinity towards any specific country or group. Popular examples of gTLDs include '.com', '.net', '.info' and '.org' domains. Absolutely anyone can buy and register a gTLD domain name from an ICANN-authorized domain registrar on a first-come first-served basis, you don't need to show any proof while registering since there's nothing of special relevance to gTLDs. They are, one can say, the most common commodities of the domain name space.

ccTLD

This is a special type of top-level domain, ccTLD or country code top-level domain, meant, as you can imagine, for identifying websites or organizations from specific countries. Where we saw no restriction on the size of gTLDs above, rules specific to ccTLDs dictate them to be no longer than two characters long. You can see it in examples such as 'nike.in' (Nike's India operations) or 'google.cn' (Google's homepage for people from China).

The rules and requirements for purchasing a ccTLD differ from one country or region to another: for e.g., anyone can just buy a '.in' domain name without showing any proof, whereas '.de' and '.ca' registrations for the German and Canadian ccTLD, respectively, require you to prove you belong to those nations.

GeoTLD

A geographic top-level domain or GeoTLD is slightly similar in concept to the above-mentioned ccTLD. Just as ccTLDs are specific to countries, GeoTLDs are specific to broader (yet specific) geographies. For example, '.asia' is an ideal example of a GeoTLD. So is the '.eu' for European Union – although '.eu' is classified as a ccTLD, it technically doesn't correspond to any particular country within the European Union. A whole list of GeoTLDs have been proposed by several domain registrars to ICANN, awaiting its approval.

sTLD

An interesting sub-set of TLDs, an sTLD or sponsored top-level domain is accredited towards certain private or government organizations which fund them. They can restrict and control the entry of new domains registered under these special digital zones. Think of them, as private, invitation-only clubs.

Prominent examples of sTLDs include '.gov' (The United States government), '.mil' (The United States Army), '.edu' (sponsored by EDUCAUSE), '.mobi' (sponsored by dotMobi), and so on and so forth. As you can see from the examples above, all the sTLDs have a sponsor or a group backing it, but some, such as '.edu' are for non-profit use. 

We bust some common jargon prevalent in the domain name and domain registration business.

1) Cybersquatting

It's a known evil of being able to register a domain with no strings attached. Cybersquatting is the practice of blocking (or buying) a domain name including the exact copyrighted characters of a company or organization (which doesn't have a website) with the motive of profiting from its resale. Many a times cybersquatters don't even have a website on the blocked domain name.

2) Domain parking

Domain parking is the practice of registering a domain name but not developing it. For example, if you buy a domain name but don't use it for hosting a website, sending receiving emails, etc., then you're merely parking a domain for future use. Domain parking done for the wrong reasons actually classifies as cybersquatting.

3) Internationalized Domain Name(IDN)

Internationalized domain name contains at least one 'label' (refer to TLD paragraph) of the domain that has non-English, international script. For example, the presence of Arabic or Chinese characters in Arabic or Chinese websites – people not conversant with these languages won't be able to read these domain names.

4) WHOIS

Simply put, WHOIS is a collection of database entries and also the means to access them over the Internet. If you need to find any information about who owns a specific domain name, WHOIS lets you access that information.